Ukkonen's suffix tree algorithm in plain English?

I feel a bit thick at this point. I've spent days trying to fully wrap my head around suffix tree construction, but because I don't have a mathematical background, many of the explanations elude me as they start to make excessive use of mathematical symbology. The closest to a good explanation that I've found is *Fast String Searching With Suffix Trees*, but he glosses over various points and some aspects of the algorithm remain unclear.

A step-by-step explanation of this algorithm here on Stack Overflow would be invaluable for many others besides me, I'm sure.

For reference, here's Ukkonen's paper on the algorithm: http://www.cs.helsinki.fi/u/ukkonen/SuffixT1withFigs.pdf

My basic understanding, so far:

- I need to iterate through each prefix P of a given string T
- I need to iterate through each suffix S in prefix P and add that to tree
- To add suffix S to the tree, I need to iterate through each character in S, with the iterations consisting of either walking down an existing branch that starts with the same set of characters C in S and potentially splitting an edge into descendent nodes when I reach a differing character in the suffix, OR if there was no matching edge to walk down. When no matching edge is found to walk down for C, a new leaf edge is created for C.

The basic algorithm appears to be $O(n^2)$, as is pointed out in most explanations, as we need to step through all of the prefixes, then we need to step through each of the suffixes for each prefix. Ukkonen's algorithm is apparently unique because of the suffix pointer technique he uses, though I think *that* is what I'm having trouble understanding.

I'm also having trouble understanding:

- exactly when and how the "active point" is assigned, used and changed
- what is going on with the canonization aspect of the algorithm
- Why the implementations I've seen need to "fix" bounding variables that they are using

EDIT (April 13, 2012)

Here is the completed source code that I've written and output based on jogojapan's answer below. The code outputs a detailed description and text-based diagram of the steps it takes as it builds the tree. It is a first version and could probably do with optimization and so forth, but it works, which is the main thing.

[Redacted URL, see updated link below]

EDIT (April 15, 2012)

The source code has been completely rewritten from scratch and now not only works correctly, but it supports automatic canonization and renders a nicer looking text graph of the output. Source code and sample output is at:

> **https://gist.github.com/2373868**

algorithm search suffix-tree

edited Jun 10 '12 at 17:49 Peter Mortensen 11.1k 15 76 109

asked Feb 26 '12 at 11:30 Nathan Ridley 15.9k 18 88 165

protected by Mysticial Aug 15 '12 at 4:56

This question is protected to prevent "thanks!", "me too!", or spam answers by new users. To answer it, you must have earned at least 10 reputation on this site (the association bonus does not count).

2 Did you take a look at the description given in Dan Gusfield's book? I found that to be helpful. – jogojapan Feb 27 '12 at 2:10

3 The gist does not specify the license - can I change your code and republish under MIT (obviously with attributions)? – Yurik Oct 11 '12 at 4:26

1 Yep, go for your life. Consider it public domain. As mentioned by another answer on this page, there's a bug that needs fixing anyway. – Nathan Ridley Oct 11 '12 at 15:27

1 maybe this implementation will help others, goto code.google.com/p/text-indexing – cos Dec 3 '13 at 8:40

1 "Consider it public domain" is, perhaps surprisingly a very unhelpful answer. The reason is that it's actually impossible for you to place the work in the public domain. Hence your "consider it..." comment underlines the fact that the license is unclear and gives the reader reason to doubt that the status of the work is actually clear to *you*. If you would like people to be able to use your code, please specify a license for it, choose any license you like (but, unless you're a lawyer, choose a pre-existing license!) – James Youngman May 21 '16 at 19:51

15 Answers

The following is an attempt to describe the Ukkonen algorithm by first showing what it does when the string is simple (i.e. does not contain any repeated characters), and then extending it to the full algorithm.

First, a few preliminary statements.

1. What we are building, is *basically* like a search trie. So there is a root node, edges going out of it leading to new nodes, and further edges going out of those, and so forth
2. **But**: Unlike in a search trie, the edge labels are not single characters. Instead, each edge is labeled using a pair of integers `[from,to]`. These are pointers into the text. In this sense, each edge carries a string label of arbitrary length, but takes only O(1) space (two pointers).

Basic principle

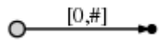
I would like to first demonstrate how to create the suffix tree of a particularly simple string, a string with no repeated characters:

```
abc
```

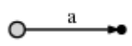
The algorithm **works in steps, from left to right**. There is **one step for every character of the string**. Each step might involve more than one individual operation, but we will see (see the final observations at the end) that the total number of operations is O(n).

So, we start from the *left*, and first insert only the single character `a` by creating an edge from the root node (on the left) to a leaf, and labeling it as `[0,#]`, which means the edge represents the substring starting at position 0 and ending at *the current end*. I use the symbol `#` to mean *the current end*, which is at position 1 (right after `a`).

So we have an initial tree, which looks like this:



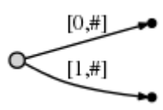
And what it means is this:



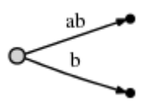
Now we progress to position 2 (right after `b`). **Our goal at each step** is to insert **all suffixes up to the current position**. We do this by

- expanding the existing `a`-edge to `ab`
- inserting one new edge for `b`

In our representation this looks like



And what it means is:

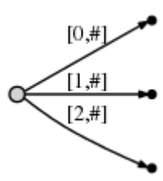


We observe two things:

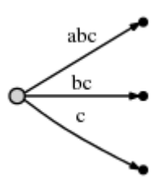
- The edge representation for `ab` is **the same** as it used to be in the initial tree: `[0,#]`. Its meaning has automatically changed because we updated the current position `#` from 1 to 2.
- Each edge consumes O(1) space, because it consists of only two pointers into the text, regardless of how many characters it represents.

Next we increment the position again and update the tree by appending a `c` to every existing edge and inserting one new edge for the new suffix `c`.

In our representation this looks like



And what it means is:



We observe:

- The tree is the correct suffix tree *up to the current position* after each step
- There are as many steps as there are characters in the text
- The amount of work in each step is O(1), because all existing edges are updated automatically by incrementing `#`, and inserting the one new edge for the final character can be done in O(1) time. Hence for a string of length n, only O(n) time is required.

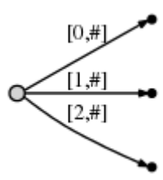
First extension: Simple repetitions

Of course this works so nicely only because our string does not contain any repetitions. We now look at a more realistic string:

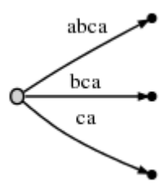
```
abcabxabcd
```

It starts with `abc` as in the previous example, then `ab` is repeated and followed by `x`, and then `abc` is repeated followed by `d`.

Steps 1 through 3: After the first 3 steps we have the tree from the previous example:



Step 4: We move `#` to position 4. This implicitly updates all existing edges to this:



and we need to insert the final suffix of the current step, `a`, at the root.

Before we do this, we introduce **two more variables** (in addition to `#`), which of course have been there all the time but we haven't used them so far:

- The **active point**, which is a triple `(active_node,active_edge,active_length)`
- The `remainder`, which is an integer indicating how many new suffixes we need to insert

The exact meaning of these two will become clear soon, but for now let's just say:

- In the simple `abc` example, the active point was always `(root,'\0x',0)`, i.e. `active_node` was the root node, `active_edge` was specified as the null character `'\0x'`, and `active_length` was zero. The effect of this was that the one new edge that we inserted in every step was inserted at the root node as a freshly created edge. We will see soon why a triple is necessary to represent this information.
- The `remainder` was always set to 1 at the beginning of each step. The meaning of this was that the number of suffixes we had to actively insert at the end of each step was 1 (always just the final character).

Now this is going to change. When we insert the current final character `a` at the root, we notice that there is already an outgoing edge starting with `a`, specifically: `abca`. Here is what we do in such a case:

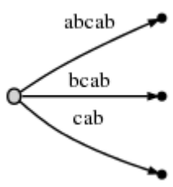
- We **do not** insert a fresh edge `[4,#]` at the root node. Instead we simply notice that the suffix `a` is already in our tree. It ends in the middle of a longer edge, but we are not bothered by that. We just leave things the way they are.
- We **set the active point** to `(root,'a',1)`. That means the active point is now somewhere in the middle of outgoing edge of the root node that starts with `a`, specifically, after position 1 on that edge. We notice that the edge is specified simply by its first character `a`. That

suffices because there can be *only one* edge starting with any particular character (confirm that this is true after reading through the entire description).

- We also increment `remainder`, so at the beginning of the next step it will be 2.

Observation: When the final **suffix we need to insert is found to exist in the tree already**, the tree itself is **not changed** at all (we only update the active point and `remainder`). The tree is then not an accurate representation of the suffix tree *up to the current position* any more, but it **contains** all suffixes (because the final suffix `a` is contained *implicitly*). Hence, apart from updating the variables (which are all of fixed length, so this is O(1)), there was **no work** done in this step.

Step 5: We update the current position `#` to 5. This automatically updates the tree to this:



And **because `remainder` is 2**, we need to insert two final suffixes of the current position: `ab` and `b`. This is basically because:

- The `a` suffix from the previous step has never been properly inserted. So it has *remained*, and since we have progressed one step, it has now grown from `a` to `ab`.
- And we need to insert the new final edge `b`.

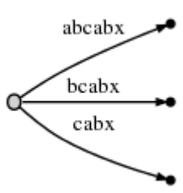
In practice this means that we go to the active point (which points to behind the `a` on what is now the `abcab` edge), and insert the current final character `b`. **But:** Again, it turns out that `b` is also already present on that same edge.

So, again, we do not change the tree. We simply:

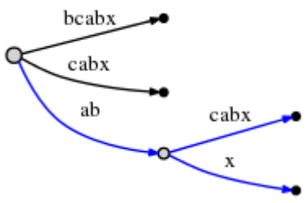
- Update the active point to `(root,'a',2)` (same node and edge as before, but now we point to behind the `b`)
- Increment the `remainder` to 3 because we still have not properly inserted the final edge from the previous step, and we don't insert the current final edge either.

To be clear: We had to insert `ab` and `b` in the current step, but because `ab` was already found, we updated the active point and did not even attempt to insert `b`. Why? Because if `ab` is in the tree, **every suffix** of it (including `b`) must be in the tree, too. Perhaps only *implicitly*, but it must be there, because of the way we have built the tree so far.

We proceed to **step 6** by incrementing `#`. The tree is automatically updated to:



Because `remainder` **is 3**, we have to insert `abx`, `bx` and `x`. The active point tells us where `ab` ends, so we only need to jump there and insert the `x`. Indeed, `x` is not there yet, so we split the `abcabx` edge and insert an internal node:



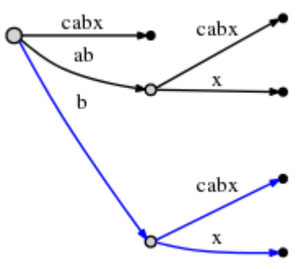
The edge representations are still pointers into the text, so splitting and inserting an internal node can be done in O(1) time.

So we have dealt with `abx` and decrement `remainder` to 2. Now we need to insert the next remaining suffix, `bx`. But before we do that we need to update the active point. The rule for this, after splitting and inserting an edge, will be called **Rule 1** below, and it applies whenever the `active_node` is root (we will learn rule 3 for other cases further below). Here is rule 1:

After an insertion from root,

- `active_node` remains root
- `active_edge` is set to the first character of the new suffix we need to insert, i.e. `b`
- `active_length` is reduced by 1

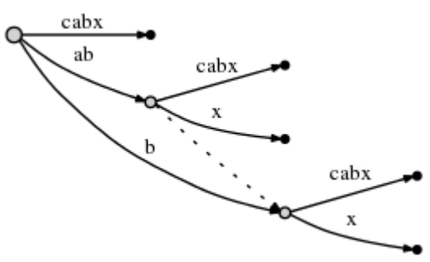
Hence, the new active-point triple `(root,'b',1)` indicates that the next insert has to be made at the `bcabx` edge, behind 1 character, i.e. behind `b`. We can identify the insertion point in O(1) time and check whether `x` is already present or not. If it was present, we would end the current step and leave everything the way it is. But `x` is not present, so we insert it by splitting the edge:



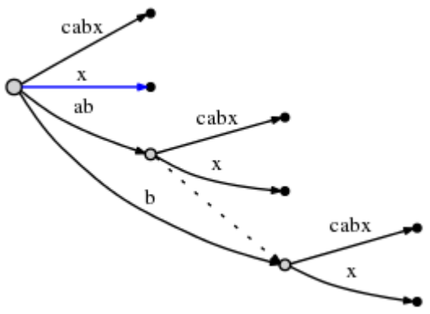
Again, this took O(1) time and we update `remainder` to 1 and the active point to `(root,'x',0)` as rule 1 states.

But there is one more thing we need to do. We'll call this **Rule 2:**

> If we split an edge and insert a new node, and if that is *not the first node* created during the current step, we connect the previously inserted node and the new node through a special pointer, a **suffix link**. We will later see why that is useful. Here is what we get, the suffix link is represented as a dotted edge:



We still need to insert the final suffix of the current step, `x`. Since the `active_length` component of the active node has fallen to 0, the final insert is made at the root directly. Since there is no outgoing edge at the root node starting with `x`, we insert a new edge:



As we can see, in the current step all remaining inserts were made.

We proceed to **step 7** by setting `#` =7, which automatically appends the next character, `a` , to all leaf edges, as always. Then we attempt to insert the new final character to the active point (the root), and find that it is there already. So we end the current step without inserting anything and update the active point to `(root,'a',1)` .

In **step 8**,, `#` =8, we append `b` , and as seen before, this only means we update the active point to `(root,'a',2)` and increment `remainder` without doing anything else, because `b` is already present. **However,** we notice (in O(1) time) that the active point is now at the end of an edge. We reflect this by re-setting it to `(node1,'\0x',0)` . Here, I use `node1` to refer to the internal node the `ab` edge ends at.

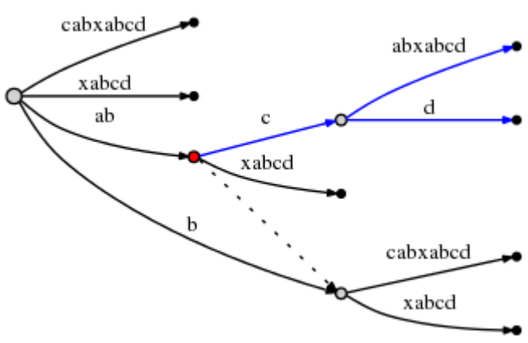
Then, in **step `#` =9**, we need to insert 'c' and this will help us to understand the final trick:

Second extension: Using suffix links

As always, the `#` update appends `c` automatically to the leaf edges and we go to the active point to see if we can insert 'c'. It turns out 'c' exists already at that edge, so we set the active point to `(node1,'c',1)` , increment `remainder` and do nothing else.

Now in **step `#` =10**, `remainder is 4` , and so we first need to insert `abcd` (which remains from 3 steps ago) by inserting `d` at the active point.

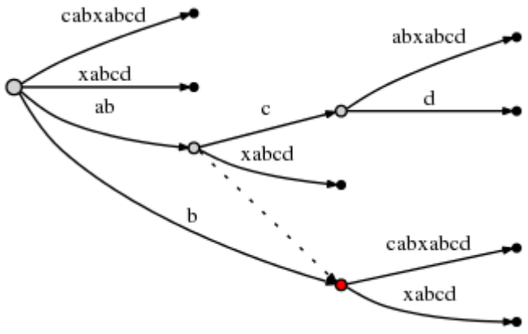
Attempting to insert `d` at the active point causes an edge split in O(1) time:



The `active_node` , from which the split was initiated, is marked in red above. Here is the final rule, **Rule 3:**

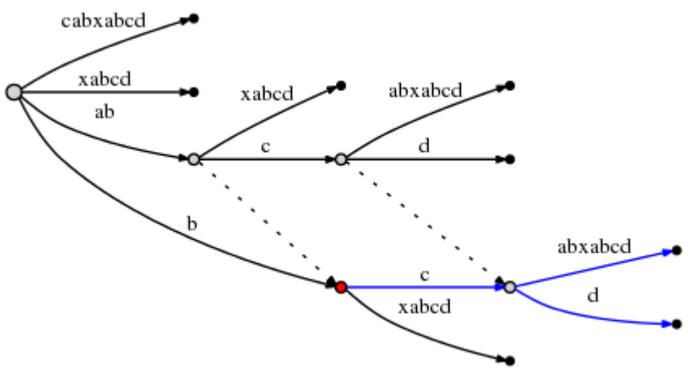
> After splitting an edge from an `active_node` that is not the root node, we follow the suffix link going out of that node, if there is any, and reset the `active_node` to the node it points to. If there is no suffix link, we set the `active_node` to the root. `active_edge` and `active_length` remain unchanged.

So the active point is now `(node2,'c',1)` , and `node2` is marked in red below:



Since the insertion of `abcd` is complete, we decrement `remainder` to 3 and consider the next remaining suffix of the current step, `bcd` . Rule 3 has set the active point to just the right node and edge so inserting `bcd` can be done by simply inserting its final character `d` at the active point.

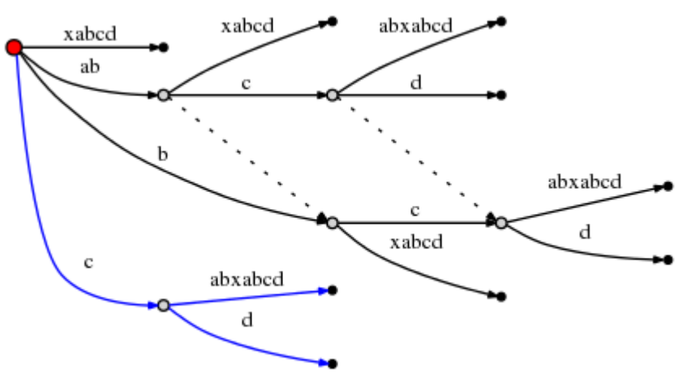
Doing this causes another edge split, and **because of rule 2**, we must create a suffix link from the previously inserted node to the new one:



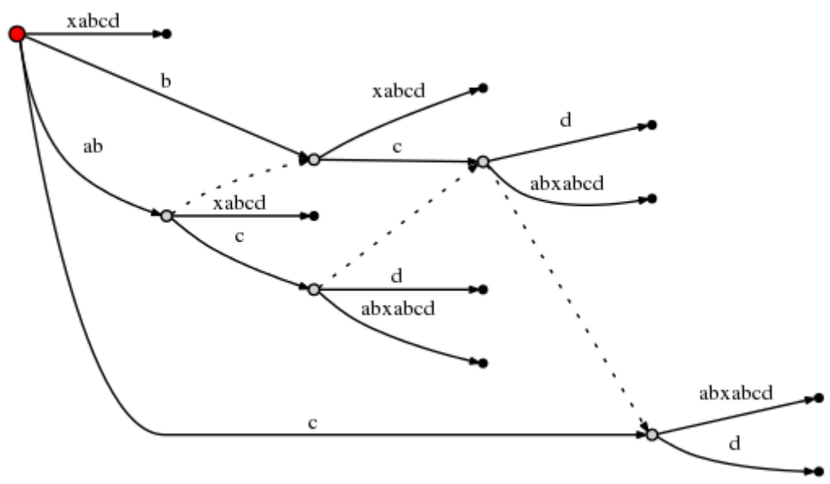
We observe: Suffix links enable us to reset the active point so we can make the next *remaining insert* at O(1) effort. Look at the graph above to confirm that indeed node at label `ab` is linked to the node at `b` (its suffix), and the node at `abc` is linked to `bc` .

The current step is not finished yet. `remainder` is now 2, and we need to follow rule 3 to reset the active point again. Since the current `active_node` (red above) has no suffix link, we reset to root. The active point is now `(root,'c',1)` .

Hence the next insert occurs at the one outgoing edge of the root node whose label starts with `c` : `cabxabcd` , behind the first character, i.e. behind `c` . This causes another split:

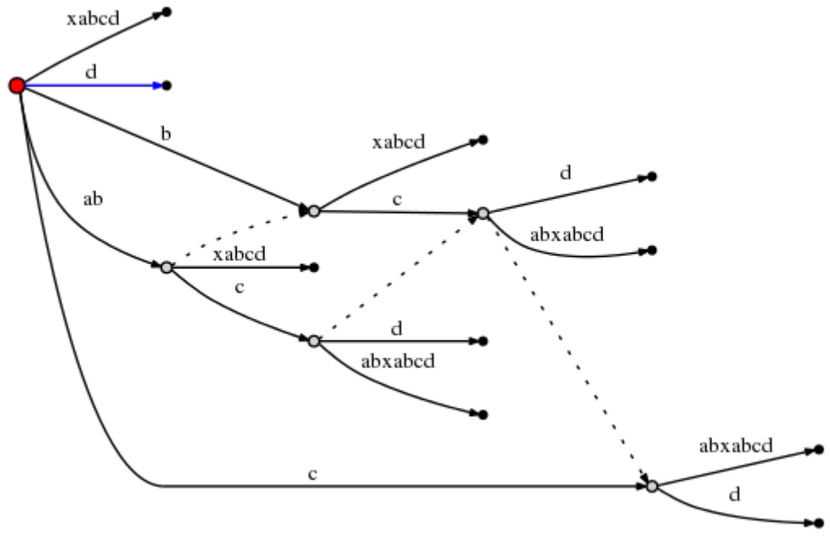


And since this involves the creation of a new internal node,we follow rule 2 and set a new suffix link from the previously created internal node:



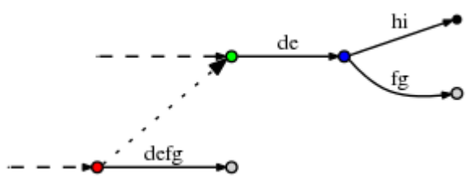
(I am using Graphviz Dot for these little graphs. The new suffix link caused dot to re-arrange the existing edges, so check carefully to confirm that the only thing that was inserted above is a new suffix link.)

With this, `remainder` can be set to 1 and since the `active_node` is root, we use rule 1 to update the active point to `(root,'d',0)`. This means the final insert of the current step is to insert a single `d` at root:



That was the final step and we are done. There are number of **final observations**, though:

- In each step we move `#` forward by 1 position. This automatically updates all leaf nodes in O(1) time.
- But it does not deal with a) any suffixes *remaining* from previous steps, and b) with the one final character of the current step.
- `remainder` tells us how many additional inserts we need to make. These inserts correspond one-to-one to the final suffixes of the string that ends at the current position `#`. We consider one after the other and make the insert. **Important:** Each insert is done in O(1) time since the active point tells us exactly where to go, and we need to add only one single character at the active point. Why? Because the other characters are *contained implicitly* (otherwise the active point would not be where it is).
- After each such insert, we decrement `remainder` and follow the suffix link if there is any. If not we go to root (rule 3). If we are at root already, we modify the active point using rule 1. In any case, it takes only O(1) time.
- If, during one of these inserts, we find that the character we want to insert is already there, we don't do anything and end the current step, even if `remainder` >0. The reason is that any inserts that remain will be suffixes of the one we just tried to make. Hence they are all *implicit* in the current tree. The fact that `remainder` >0 makes sure we deal with the remaining suffixes later.
- What if at the end of the algorithm `remainder` >0? This will be the case whenever the end of the text is a substring that occurred somewhere before. In that case we must append one extra character at the end of the string that has not occurred before. In the literature, usually the dollar sign `$` is used as a symbol for that. **Why does that matter?** --> If later we use the completed suffix tree to search for suffixes, we must accept matches only if they *end at a leaf*. Otherwise we would get a lot of spurious matches, because there are *many* strings *implicitly* contained in the tree that are not actual suffixes of the main string. Forcing `remainder` to be 0 at the end is essentially a way to ensure that all suffixes end at a leaf node. **However,** if we want to use the tree to search for *general substrings*, not only *suffixes* of the main string, this final step is indeed not required, as suggested by the OP's comment below.
- So what is the complexity of the entire algorithm? If the text is n characters in length, there are obviously n steps (or n+1 if we add the dollar sign). In each step we either do nothing (other than updating the variables), or we make `remainder` inserts, each taking O(1) time. Since `remainder` indicates how many times we have done nothing in previous steps, and is decremented for every insert that we make now, the total number of times we do something is exactly n (or n+1). Hence, the total complexity is O(n).
- However, there is one small thing that I did not properly explain: It can happen that we follow a suffix link, update the active point, and then find that its `active_length` component does not work well with the new `active_node`. For example, consider a situation like this:



(The dashed lines indicate the rest of the tree. The dotted line is a suffix link.)

Now let the active point be `(red,'d',3)`, so it points to the place behind the `f` on the `defg` edge. Now assume we made the necessary updates and now follow the suffix link to update the active point according to rule 3. The new active point is `(green,'d',3)`. However, the `d`-edge going out of the green node is `de`, so it has only 2 characters. In order to find the correct active point, we obviously need to follow that edge to the blue node and reset to `(blue,'f',1)`.

In a particularly bad case, the `active_length` could be as large as `remainder`, which can be as large as n. And it might very well happen that to find the correct active point, we need not only jump over one internal node, but perhaps many, up to n in the worst case. Does that mean the algorithm has a hidden $O(n^2)$ complexity, because in each step `remainder` is generally O(n), and the post-adjustments to the active node after following a suffix link could be O(n), too?

No. The reason is that if indeed we have to adjust the active point (e.g. from green to blue as above), that brings us to a new node that has its own suffix link, and `active_length` will be reduced. As we follow down the chain of suffix links we make the remaining inserts, `active_length` can only decrease, and the number of active-point adjustments we can make on the way can't be larger than `active_length` at any given time. Since `active_length` can never be

larger than `remainder`, and `remainder` is O(n) not only in every single step, but the total sum of increments ever made to `remainder` over the course of the entire process is O(n) too, the number of active point adjustments is also bounded by O(n).

edited May 23 '13 at 20:38 — Justin Poliey — **15.3k** 5 30 45

answered Mar 1 '12 at 9:13 — jogojapan — **43.1k** 7 65 92

49 Sorry this ended up a little longer than I hoped. And I realize it explains an number of trivial things we all know, while the difficult parts might still not be perfectly clear. Let's edit it into shape together. – jogojapan Mar 1 '12 at 9:14

50 And I should add that this is *not* based on the description found in Dan Gusfield's book. It's a new attempt at describing the algorithm by first considering a string with no repetitions and then discussing how repetitions are handled. I hoped that would be more intuitive. – jogojapan Mar 1 '12 at 9:16

147 Oh, to everyone else, vote this man up! Such answers deserve a significant number of votes! – Nathan Ridley Mar 1 '12 at 14:42

54 This is why I love stack overflow. Epic answer. – Tom Anderson Apr 11 '12 at 23:15

41 Outstanding answer. It helps that the question was actually interesting and well presented and hence would encourage a quality answer, but this is above and beyond. Would love to see more of this on SO. Kudos @jogojapan. – OJ. Apr 11 '12 at 23:28

I tried to implement the Suffix Tree with the approach given in jogojapan's answer, but it didn't work for some cases due to wording used for the rules. Moreover, I've mentioned that nobody managed to implement an absolutely correct suffix tree using this approach. Below I will write an "overview" of jogojapan's answer with some modifications to the rules. I will also describe the case when we forget to create ***important*** suffix links.

Additional variables used

1. **active point** - a triple (active_node; active_edge; active_length), showing from where we must start inserting a new suffix.
2. **remainder** - shows the number of suffixes we must add *explicitly*. For instance, if our word is 'abcaabca', and remainder = 3, it means we must process 3 last suffixes: **bca**, **ca** and **a**.

Let's use a concept of an **internal node** - all the nodes, except the *root* and the *leafs* are **internal nodes**.

Observation 1

When the final suffix we need to insert is found to exist in the tree already, the tree itself is not changed at all (we only update the `active point` and `remainder`).

Observation 2

If at some point `active_length` is greater or equal to the length of current edge (`edge_length`), we move our `active point` down until `edge_length` is strictly greater than `active_length`.

Now, let's redefine the rules:

Rule 1

> If after an insertion from the *active node* = *root*, the *active length* is greater than 0, then:
>
> 1. *active node* is not changed
> 2. *active length* is decremented
> 3. *active edge* is shifted right (to the first character of the next suffix we must insert)

Rule 2

> If we create a new *internal node* **OR** make an inserter from an *internal node*, and this is not the first **SUCH** *internal node* at current step, then we link the previous **SUCH** node with **THIS** one through a *suffix link*.

This definition of the `Rule 2` is different from jogojapan', as here we take into account not only the *newly created* internal nodes, but also the internal nodes, from which we make an insertion.

Rule 3

> After an insert from the *active node* which is not the *root* node, we must follow the suffix link and set the *active node* to the node it points to. If there is no a suffix link, set the *active node* to the *root* node. Either way, *active edge* and *active length* stay unchanged.

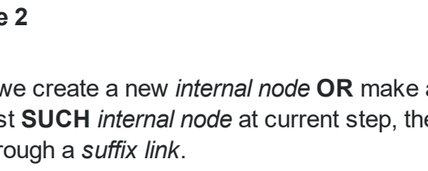
In this definition of `Rule 3` we also consider the inserts of leaf nodes (not only split-nodes).

And finally, Observation 3:

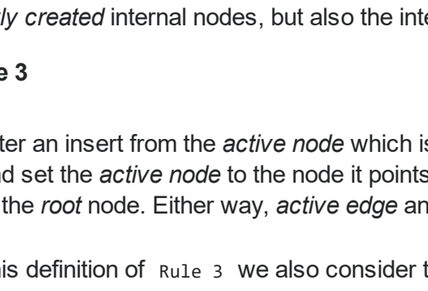
When the symbol we want to add to the tree is already on the edge, we, according to `Observation 1`, update only `active point` and `remainder`, leaving the tree unchanged. **BUT** if there is an *internal node* marked as *needing suffix link*, we must connect that node with our current `active node` through a suffix link.

Let's look at the example of a suffix tree for **cdddcdc** if we add a suffix link in such case and if we don't:

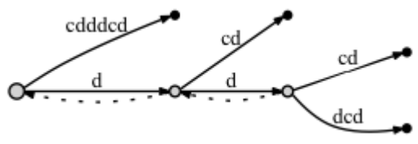
1. If we **DON'T** connect the nodes through a suffix link:
   - before adding the last letter **c**:



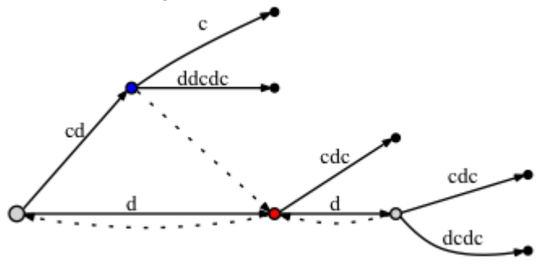
   - after adding the last letter **c**:



2. If we **DO** connect the nodes through a suffix link:
   - before adding the last letter **c**:



- after adding the last letter **c**:



Seems like there is no significant difference: in the second case there are two more suffix links. But these suffix links are *correct*, and one of them - from the blue node to the red one - is very **important** for our approach with **active point**. The problem is that if we don't put a suffix link here, later, when we add some new letters to the tree, we might omit adding some nodes to the tree due to the `Rule 3`, because, according to it, if there's no a suffix link, then we must put the `active_node` to the root.

When we were adding the last letter to the tree, the red node had **already existed** before we made an insert from the blue node (the edge labled **'c'**). As there was an insert from the blue node, we mark it as *needing a suffix link*. Then, relying on the **active point** approach, the `active node` was set to the red node. But we don't make an insert from the red node, as the letter **'c'** is already on the edge. Does it mean that the blue node must be left without a suffix link? No, we must connect the blue node with the red one through a suffix link. Why is it correct? Because the **active point** approach guarantees that we get to a right place, i.e., to the next place where we must process an insert of a **shorter** suffix.

Finally, here are my implementations of the Suffix Tree:

1. Java
2. C++

Hope that this "overview" combined with jogojapan's detailed answer will help somebody to implement his own Suffix Tree.

edited Oct 10 '16 at 23:15

answered Jan 29 '13 at 9:57

makagonov
1,122 7 10

1 Thanks very much and +1 for you effort. I am sure you are right.. although I don't have the time to think about the details right away. I'll check later and possibly modify my answer then as well. – jogojapan Jan 30 '13 at 1:30

Thanks so much, it really helped. Though, could you be more specific on Observation 3? For instance, giving the diagrams of the 2 steps that introduce the new suffix link. Is the node linked the active node? (as we don't actually insert the 2nd node) – dyesdyes Jun 8 '14 at 16:15

@makagonov Hey can you help me build suffix tree for your string "cdddcdc" I am bit confused doing so (the starting steps). – tariq zafar Nov 10 '14 at 6:20

1 As for rule 3, a smart way is to set the suffix link of root to root itself, and (in default) set the suffix link of every node to root. Thus we can avoid the conditioning and just follow the suffix link. – sqd May 1 '15 at 9:41

Great explanation! Just one thing I'd like to add. I am still trying to understand, but as far as I understood, this explanation skips one edge case: "abca". The case where the last digit was repeated in the string before. In that case we need to append an end character to the string, sth like '$'. Check the 5th paragraph in cise.ufl.edu/~sahni/dsaaj/enrich/c16/suffix.htm – elif Dec 21 '15 at 7:54

|

I have had quite a lot problems to implement this data structure myself. In the end I found this article and managed to implement it. A great plus for it is that you have a step-by-step example for a quite long string so you get to see what you should do. Please take a look at the article and I will be more then happy to give tips where needed. I am hesitant to gove yet another full-blown explanation as there are alredy quite a few round the internet.

answered Feb 26 '12 at 11:59

Ivaylo Strandjev
50.5k 10 68 124

1 Yeah I've seen that one but again there's a lot of text and symbology that I am having trouble with. I think I need an illustrated picture book to understand it properly :P – Nathan Ridley Feb 26 '12 at 12:03

1 There are tons of pictures in this article. Please be patient and try to read it. At first the author tries to explain why is the suffix tree good and then shows in great detail how to build it(pictures included on each step). Still there are a few things you can't evade - you need to know what is suffix and compressed trie at least. – Ivaylo Strandjev Feb 26 '12 at 12:06

I intend to write my own blog post once I have my head wrapped around this, and I'm going to illustrate every aspect of the algorithm, step by step. – Nathan Ridley Feb 26 '12 at 12:07

I would happily help you with that. Also should you have any question on a given part of the algorithm I will be glad to answer – Ivaylo Strandjev Feb 26 '12 at 12:10

1 Ok I just re-read the article and the bit that lost me is when he started talking about "proper" suffixes without explaining what he meant by "proper". – Nathan Ridley Feb 26 '12 at 12:12

|

My intuition is as follows:

After k iterations of the main loop you have constructed a suffix tree which contains all suffixes of the complete string that start in the first k characters.

At the start, this means the suffix tree contains a single root node that represents the entire string (this is the only suffix that starts at 0).

After len(string) iterations you have a suffix tree that contains all suffixes.

During the loop the key is the active point. My guess is that this represents the deepest point in the suffix tree that corresponds to a proper suffix of the first k characters of the string. (I think proper means that the suffix cannot be the entire string.)

For example, suppose you have seen characters 'abcabc'. The active point would represent the point in the tree corresponding to the suffix 'abc'.

The active point is represented by (origin,first,last). This means that you are currently at the point in the tree that you get to by starting at node origin and then feeding in the characters in string[first:last]

When you add a new character you look to see whether the active point is still in the existing tree. If it is then you are done. Otherwise you need to add a new node to the suffix tree at the active point, fallback to the next shortest match, and check again.

Note 1: The suffix pointers give a link to the next shortest match for each node.

Note 2: When you add a new node and fallback you add a new suffix pointer for the new node. The destination for this suffix pointer will be the node at the shortened active point. This node will either already exist, or be created on the next iteration of this fallback loop.

Note 3: The canonization part simply saves time in checking the active point. For example, suppose you always used origin=0, and just changed first and last. To check the active point you would have to follow the suffix tree each time along all the intermediate nodes. It makes sense to cache the result of following this path by recording just the distance from the last node.

Can you give a code example of what you mean by "fix" bounding variables?

Health warning: I also found this algorithm particularly hard to understand so please realise that this intuition is likely to be incorrect in all important details...

edited Feb 26 '12 at 20:41

answered Feb 26 '12 at 20:16

Peter de Rivaz
22k 4 21 50

One of the academic papers defines "proper" as meaning the "proper suffix" of a string doesn't contain its first character. Sometimes you call a whole substring a "suffix", but when defining the algorithm the terms "string" and "substring" and "suffix" get thrown around liberally and sometimes you need to be very clear what you mean by "suffix", so the term "proper suffix" excludes calling the whole thing a suffix. So a suffix substring of a string can be any legitimate substring and can have a proper suffix that isn't the same suffix. Because logic. – Blair Houghton May 23 '16 at 16:26

Hi i have tried to implement the above explained implementation in ruby , please check it out. it seems to work fine.

the only difference in the implementation is that , i have tried to use the edge object instead of just using symbols.

its also present at https://gist.github.com/suchitpuri/9304856

```
    require 'pry'


class Edge
    attr_accessor :data , :edges , :suffix_link
    def initialize data
        @data = data
        @edges = []
        @suffix_link = nil
    end

    def find_edge element
        self.edges.each do |edge|
            return edge if edge.data.start_with? element
        end
        return nil
    end
end

class SuffixTrees
    attr_accessor :root , :active_point , :remainder , :pending_prefixes ,
:last_split_edge , :remainder

    def initialize
        @root = Edge.new nil
        @active_point = { active_node: @root , active_edge: nil , active_length: 0}
        @remainder = 0
        @pending_prefixes = []
        @last_split_edge = nil
        @remainder = 1
    end

    def build string
        string.split("").each_with_index do |element , index|


            add_to_edges @root , element

            update_pending_prefix element
            add_pending_elements_to_tree element
            active_length = @active_point[:active_length]

            # if(@active_point[:active_edge] && @active_point[:active_edge].data &&
@active_point[:active_edge].data[0..active_length-1] ==
@active_point[:active_edge].data[active_length..@active_point[:active_edge].data.length-
1])
            #   @active_point[:active_edge].data =
@active_point[:active_edge].data[0..active_length-1]
            #   @active_point[:active_edge].edges <<
Edge.new(@active_point[:active_edge].data)
            # end

            if(@active_point[:active_edge] && @active_point[:active_edge].data &&
@active_point[:active_edge].data.length == @active_point[:active_length]  )
                @active_point[:active_node] =  @active_point[:active_edge]
                @active_point[:active_edge] =
@active_point[:active_node].find_edge(element[0])
                @active_point[:active_length] = 0
            end
        end
    end

    def add_pending_elements_to_tree element

        to_be_deleted = []
        update_active_length = false
        # binding.pry
        if( @active_point[:active_node].find_edge(element[0]) != nil)
            @active_point[:active_length] = @active_point[:active_length] + 1
            @active_point[:active_edge] =
@active_point[:active_node].find_edge(element[0]) if @active_point[:active_edge] == nil
            @remainder = @remainder + 1
            return
        end



        @pending_prefixes.each_with_index do |pending_prefix , index|

            # binding.pry

            if @active_point[:active_edge] == nil and
@active_point[:active_node].find_edge(element[0]) == nil

                @active_point[:active_node].edges << Edge.new(element)

            else

                @active_point[:active_edge] = node.find_edge(element[0]) if
@active_point[:active_edge]  == nil

                data = @active_point[:active_edge].data
                data = data.split("")

                location = @active_point[:active_length]


                # binding.pry
                if(data[0..location].join == pending_prefix or
```

```
    @active_point[:active_node].find_edge(element) != nil )


                else #tree split
                    split_edge data , index , element
                end

            end
        end
    end



    def update_pending_prefix element
        if @active_point[:active_edge] == nil
            @pending_prefixes = [element]
            return

        end

        @pending_prefixes = []

        length = @active_point[:active_edge].data.length
        data = @active_point[:active_edge].data
        @remainder.times do |ctr|
                @pending_prefixes << data[-(ctr+1)..data.length-1]
        end

        @pending_prefixes.reverse!

    end

    def split_edge data , index , element
        location = @active_point[:active_length]
        old_edges = []
        internal_node = (@active_point[:active_edge].edges != nil)

        if (internal_node)
            old_edges = @active_point[:active_edge].edges
            @active_point[:active_edge].edges = []
        end

        @active_point[:active_edge].data = data[0..location-1].join
        @active_point[:active_edge].edges << Edge.new(data[location..data.size].join)


        if internal_node
            @active_point[:active_edge].edges << Edge.new(element)
        else
            @active_point[:active_edge].edges << Edge.new(data.last)
        end

        if internal_node
            @active_point[:active_edge].edges[0].edges = old_edges
        end


        #setup the suffix link
        if @last_split_edge != nil and @last_split_edge.data.end_with?
@active_point[:active_edge].data

            @last_split_edge.suffix_link = @active_point[:active_edge]
        end

        @last_split_edge = @active_point[:active_edge]

        update_active_point index

    end


    def update_active_point index
        if(@active_point[:active_node] == @root)
            @active_point[:active_length] = @active_point[:active_length] - 1
            @remainder = @remainder - 1
            @active_point[:active_edge] =
@active_point[:active_node].find_edge(@pending_prefixes.first[index+1])
        else
            if @active_point[:active_node].suffix_link != nil
                @active_point[:active_node] = @active_point[:active_node].suffix_link
            else
                @active_point[:active_node] = @root
            end
            @active_point[:active_edge] =
@active_point[:active_node].find_edge(@active_point[:active_edge].data[0])
            @remainder = @remainder - 1
        end
    end

    def add_to_edges root , element
        return if root == nil
        root.data = root.data + element if(root.data and root.edges.size == 0)
        root.edges.each do |edge|
            add_to_edges edge , element
        end
    end
end

suffix_tree = SuffixTrees.new
suffix_tree.build("abcabxabcd")
binding.pry
```

edited Mar 3 '14 at 3:32

answered Mar 2 '14 at 10:54

Suchit Puri

305 1 7

@jogojapan you brought awesome explanation and visualisation. But as @makagonov mentioned it's missing some rules regarding setting suffix links. It's visible in nice way when going step by step on http://brenden.github.io/ukkonen-animation/ through word 'aabaaabb'. When you go from step 10 to step 11, there is no suffix link from node 5 to node 2 but active point suddenly moves there.

@makagonov since I live in Java world I also tried to follow your implementation to grasp ST building workflow but it was hard to me because of:

- combining edges with nodes
- using index pointers instead of references
- breaks statements;
- continue statements;

So I ended up with such implementation in Java which I hope reflects all steps in clearer way and will reduce learning time for other Java people:

```
import java.util.Arrays;
import java.util.HashMap;
import java.util.Map;

public class ST {

  public class Node {
    private final int id;
    private final Map<Character, Edge> edges;
    private Node slink;
```

```
        public Node(final int id) {
            this.id = id;
            this.edges = new HashMap<>();
        }

        public void setSlink(final Node slink) {
            this.slink = slink;
        }

        public Map<Character, Edge> getEdges() {
            return this.edges;
        }

        public Node getSlink() {
            return this.slink;
        }

        public String toString(final String word) {
            return new StringBuilder()
                    .append("{")
                    .append("\"id\"")
                    .append(":")
                    .append(this.id)
                    .append(",")
                    .append("\"slink\"")
                    .append(":")
                    .append(this.slink != null ? this.slink.id : null)
                    .append(",")
                    .append("\"edges\"")
                    .append(":")
                    .append(edgesToString(word))
                    .append("}")
                    .toString();
        }

      private StringBuilder edgesToString(final String word) {
          final StringBuilder edgesStringBuilder = new StringBuilder();
          edgesStringBuilder.append("{");
          for(final Map.Entry<Character, Edge> entry : this.edges.entrySet()) {
              edgesStringBuilder.append("\"")
                      .append(entry.getKey())
                      .append("\"")
                      .append(":")
                      .append(entry.getValue().toString(word))
                      .append(",");
          }
          if(!this.edges.isEmpty()) {
              edgesStringBuilder.deleteCharAt(edgesStringBuilder.length() - 1);
          }
          edgesStringBuilder.append("}");
          return edgesStringBuilder;
      }

      public boolean contains(final String word, final String suffix) {
          return !suffix.isEmpty()
                  && this.edges.containsKey(suffix.charAt(0))
                  && this.edges.get(suffix.charAt(0)).contains(word, suffix);
      }
    }

    public class Edge {
      private final int from;
      private final int to;
      private final Node next;

      public Edge(final int from, final int to, final Node next) {
          this.from = from;
          this.to = to;
          this.next = next;
      }

      public int getFrom() {
          return this.from;
      }

      public int getTo() {
          return this.to;
      }

      public Node getNext() {
          return this.next;
      }

      public int getLength() {
          return this.to - this.from;
      }

      public String toString(final String word) {
          return new StringBuilder()
                  .append("{")
                  .append("\"content\"")
                  .append(":")
                  .append("\"")
                  .append(word.substring(this.from, this.to))
                  .append("\"")
                  .append(",")
                  .append("\"next\"")
                  .append(":")
                  .append(this.next != null ? this.next.toString(word) : null)
                  .append("}")
                  .toString();
      }

      public boolean contains(final String word, final String suffix) {
          if(this.next == null) {
              return word.substring(this.from, this.to).equals(suffix);
          }
          return suffix.startsWith(word.substring(this.from,
                  this.to)) && this.next.contains(word, suffix.substring(this.to -
    this.from));
      }
    }

    public class ActivePoint {
      private final Node activeNode;
      private final Character activeEdgeFirstCharacter;
      private final int activeLength;

      public ActivePoint(final Node activeNode,
                         final Character activeEdgeFirstCharacter,
                         final int activeLength) {
          this.activeNode = activeNode;
          this.activeEdgeFirstCharacter = activeEdgeFirstCharacter;
          this.activeLength = activeLength;
      }

      private Edge getActiveEdge() {
          return this.activeNode.getEdges().get(this.activeEdgeFirstCharacter);
      }

      public boolean pointsToActiveNode() {
          return this.activeLength == 0;
      }

      public boolean activeNodeIs(final Node node) {
          return this.activeNode == node;
      }

      public boolean activeNodeHasEdgeStartingWith(final char character) {
          return this.activeNode.getEdges().containsKey(character);
      }

      public boolean activeNodeHasSlink() {
          return this.activeNode.getSlink() != null;
```

```
        }

        public boolean pointsToOnActiveEdge(final String word, final char character) {
            return word.charAt(this.getActiveEdge().getFrom() + this.activeLength) ==
    character;
        }

        public boolean pointsToTheEndOfActiveEdge() {
            return this.getActiveEdge().getLength() == this.activeLength;
        }

        public boolean pointsAfterTheEndOfActiveEdge() {
            return this.getActiveEdge().getLength() < this.activeLength;
        }

        public ActivePoint moveToEdgeStartingWithAndByOne(final char character) {
            return new ActivePoint(this.activeNode, character, 1);
        }

        public ActivePoint moveToNextNodeOfActiveEdge() {
            return new ActivePoint(this.getActiveEdge().getNext(), null, 0);
        }

        public ActivePoint moveToSlink() {
            return new ActivePoint(this.activeNode.getSlink(),
                    this.activeEdgeFirstCharacter,
                    this.activeLength);
        }

        public ActivePoint moveTo(final Node node) {
            return new ActivePoint(node, this.activeEdgeFirstCharacter, this.activeLength);
        }

        public ActivePoint moveByOneCharacter() {
            return new ActivePoint(this.activeNode,
                    this.activeEdgeFirstCharacter,
                    this.activeLength + 1);
        }

        public ActivePoint moveToEdgeStartingWithAndByActiveLengthMinusOne(final Node node,
                                                                           final char
    character) {
            return new ActivePoint(node, character, this.activeLength - 1);
        }

        public ActivePoint moveToNextNodeOfActiveEdge(final String word, final int index) {
            return new ActivePoint(this.getActiveEdge().getNext(),
                    word.charAt(index - this.activeLength + this.getActiveEdge().getLength()),
                    this.activeLength - this.getActiveEdge().getLength());
        }

        public void addEdgeToActiveNode(final char character, final Edge edge) {
            this.activeNode.getEdges().put(character, edge);
        }

        public void splitActiveEdge(final String word,
                                    final Node nodeToAdd,
                                    final int index,
                                    final char character) {
            final Edge activeEdgeToSplit = this.getActiveEdge();
            final Edge splittedEdge = new Edge(activeEdgeToSplit.getFrom(),
                    activeEdgeToSplit.getFrom() + this.activeLength,
                    nodeToAdd);
            nodeToAdd.getEdges().put(word.charAt(activeEdgeToSplit.getFrom() +
    this.activeLength),
                    new Edge(activeEdgeToSplit.getFrom() + this.activeLength,
                            activeEdgeToSplit.getTo(),
                            activeEdgeToSplit.getNext()));
            nodeToAdd.getEdges().put(character, new Edge(index, word.length(), null));
            this.activeNode.getEdges().put(this.activeEdgeFirstCharacter, splittedEdge);
        }
```

```
      public Node setSlinkTo(final Node previouslyAddedNodeOrAddedEdgeNode,
                             final Node node) {
          if(previouslyAddedNodeOrAddedEdgeNode != null) {
              previouslyAddedNodeOrAddedEdgeNode.setSlink(node);
          }
          return node;
      }

      public Node setSlinkToActiveNode(final Node previouslyAddedNodeOrAddedEdgeNode) {
          return setSlinkTo(previouslyAddedNodeOrAddedEdgeNode, this.activeNode);
      }
    }

    private static int idGenerator;

    private final String word;
    private final Node root;
    private ActivePoint activePoint;
    private int remainder;

    public ST(final String word) {
      this.word = word;
      this.root = new Node(idGenerator++);
      this.activePoint = new ActivePoint(this.root, null, 0);
      this.remainder = 0;
      build();
    }

    private void build() {
      for(int i = 0; i < this.word.length(); i++) {
          add(i, this.word.charAt(i));
      }
    }

    private void add(final int index, final char character) {
      this.remainder++;
      boolean characterFoundInTheTree = false;
      Node previouslyAddedNodeOrAddedEdgeNode = null;
        while(!characterFoundInTheTree && this.remainder > 0) {
            if(this.activePoint.pointsToActiveNode()) {
                if(this.activePoint.activeNodeHasEdgeStartingWith(character)) {
                    activeNodeHasEdgeStartingWithCharacter(character,
    previouslyAddedNodeOrAddedEdgeNode);
                    characterFoundInTheTree = true;
                }
                else {
                    if(this.activePoint.activeNodeIs(this.root)) {
                        rootNodeHasNotEdgeStartingWithCharacter(index, character);
                    }
                    else {
                        previouslyAddedNodeOrAddedEdgeNode =
    internalNodeHasNotEdgeStartingWithCharacter(index,
                                character, previouslyAddedNodeOrAddedEdgeNode);
                    }
                }
            }
            else {
                if(this.activePoint.pointsToOnActiveEdge(this.word, character)) {
                    activeEdgeHasCharacter();
                    characterFoundInTheTree = true;
                }
                else {
                    if(this.activePoint.activeNodeIs(this.root)) {
                        previouslyAddedNodeOrAddedEdgeNode =
    edgeFromRootNodeHasNotCharacter(index,
                                character,
                                previouslyAddedNodeOrAddedEdgeNode);
                    }
                    else {
                        previouslyAddedNodeOrAddedEdgeNode =
    edgeFromInternalNodeHasNotCharacter(index,
                                character,
```

```
                            previouslyAddedNodeOrAddedEdgeNode);
                }
            }
        }
      }
    }

    private void activeNodeHasEdgeStartingWithCharacter(final char character,
                                                        final Node
previouslyAddedNodeOrAddedEdgeNode) {
      this.activePoint.setSlinkToActiveNode(previouslyAddedNodeOrAddedEdgeNode);
      this.activePoint = this.activePoint.moveToEdgeStartingWithAndByOne(character);
      if(this.activePoint.pointsToTheEndOfActiveEdge()) {
          this.activePoint = this.activePoint.moveToNextNodeOfActiveEdge();
      }
    }

    private void rootNodeHasNotEdgeStartingWithCharacter(final int index, final char
character) {
      this.activePoint.addEdgeToActiveNode(character, new Edge(index, this.word.length(),
null));
      this.activePoint = this.activePoint.moveTo(this.root);
      this.remainder--;
      assert this.remainder == 0;
    }

    private Node internalNodeHasNotEdgeStartingWithCharacter(final int index,
                                                             final char character,
                                                             Node
previouslyAddedNodeOrAddedEdgeNode) {
      this.activePoint.addEdgeToActiveNode(character, new Edge(index, this.word.length(),
null));
      previouslyAddedNodeOrAddedEdgeNode =
this.activePoint.setSlinkToActiveNode(previouslyAddedNodeOrAddedEdgeNode);
      if(this.activePoint.activeNodeHasSlink()) {
          this.activePoint = this.activePoint.moveToSlink();
      }
      else {
          this.activePoint = this.activePoint.moveTo(this.root);
      }
      this.remainder--;
      return previouslyAddedNodeOrAddedEdgeNode;
    }

    private void activeEdgeHasCharacter() {
      this.activePoint = this.activePoint.moveByOneCharacter();
      if(this.activePoint.pointsToTheEndOfActiveEdge()) {
          this.activePoint = this.activePoint.moveToNextNodeOfActiveEdge();
      }
    }

    private Node edgeFromRootNodeHasNotCharacter(final int index,
                                                 final char character,
                                                 Node previouslyAddedNodeOrAddedEdgeNode) {
      final Node newNode = new Node(idGenerator++);
      this.activePoint.splitActiveEdge(this.word, newNode, index, character);
      previouslyAddedNodeOrAddedEdgeNode =
this.activePoint.setSlinkTo(previouslyAddedNodeOrAddedEdgeNode, newNode);
      this.activePoint =
this.activePoint.moveToEdgeStartingWithAndByActiveLengthMinusOne(this.root,
              this.word.charAt(index - this.remainder + 2));
      this.activePoint = walkDown(index);
      this.remainder--;
      return previouslyAddedNodeOrAddedEdgeNode;
    }

    private Node edgeFromInternalNodeHasNotCharacter(final int index,
                                                     final char character,
                                                     Node previouslyAddedNodeOrAddedEdgeNode)
{
      final Node newNode = new Node(idGenerator++);
      this.activePoint.splitActiveEdge(this.word, newNode, index, character);
      previouslyAddedNodeOrAddedEdgeNode =
this.activePoint.setSlinkTo(previouslyAddedNodeOrAddedEdgeNode, newNode);
      if(this.activePoint.activeNodeHasSlink()) {
          this.activePoint = this.activePoint.moveToSlink();
      }
      else {
          this.activePoint = this.activePoint.moveTo(this.root);
      }
      this.activePoint = walkDown(index);
      this.remainder--;
      return previouslyAddedNodeOrAddedEdgeNode;
    }

    private ActivePoint walkDown(final int index) {
      while(!this.activePoint.pointsToActiveNode()
              && (this.activePoint.pointsToTheEndOfActiveEdge() ||
this.activePoint.pointsAfterTheEndOfActiveEdge())) {
          if(this.activePoint.pointsAfterTheEndOfActiveEdge()) {
              this.activePoint = this.activePoint.moveToNextNodeOfActiveEdge(this.word,
index);
          }
          else {
              this.activePoint = this.activePoint.moveToNextNodeOfActiveEdge();
          }
      }
      return this.activePoint;
    }

    public String toString(final String word) {
      return this.root.toString(word);
    }

    public boolean contains(final String suffix) {
      return this.root.contains(this.word, suffix);
    }

    public static void main(final String[] args) {
      final String[] words = {
              "abcabcabc$",
              "abc$",
              "abcabxabcd$",
              "abcabxabda$",
              "abcabxad$",
              "aabaaabb$",
              "aababcabcd$",
              "ababcabcd$",
              "abccba$",
              "mississipi$",
              "abacabadabacabae$",
              "abcabcd$",
              "00132220$"
      };
      Arrays.stream(words).forEach(word -> {
          System.out.println("Building suffix tree for word: " + word);
          final ST suffixTree = new ST(word);
          System.out.println("Suffix tree: " + suffixTree.toString(word));
          for(int i = 0; i < word.length() - 1; i++) {
              assert suffixTree.contains(word.substring(i)) : word.substring(i);
          }
      });
    }
}
```

edited Apr 29 at 17:02

answered Apr 21 at 14:22

Kamil
171 2 7

Regarding the implementation (https://gist.github.com/2373868), when calling `tree.WriteDot()` , it

gets an error on line 340, in the getter of the `String` property, because the length provided to the `Substring` method is too big (off by one).

Removing the `+1` fixes it, since you are only using this property inside the `WriteDotGraph` method of the `Edge` class, but there might be a bug somewhere.

deleted by ThiefMaster ♦ Aug 13 '12 at 21:15

answered Jul 6 '12 at 18:48
bb01234
305 2 13

I've found it simplier to understand a kart-trie. A kart-trie is like a crit-bit, radix or patricia trie but it uses a hash key to simplify the insertion of new nodes. I've wrote an php implementation at phpclasses.org.

deleted by owner Mar 5 '12 at 2:52

answered Feb 26 '12 at 12:27
Betterdev
11k 5 19 65

I have another question in this topic, how to implement suffix tree in XML tree supposed the pre-order traversal? when to call suffix links and when to set the suffix nodes? any one could help me in the implementation?

deleted by George Stocker ♦ Jul 5 '12 at 16:00

answered Jul 5 '12 at 15:52
ÚÒíÒ ÇáËÚÇáÈí
16 3

"We set the active point to (root,'a',1). That means the active point is now somewhere in the middle of outgoing edge of the root node that starts with a, specifically, after position 1 on that edge. "

Then

"We proceed to step 7 by setting #=7, which automatically appends the next character, a, to all leaf edges, as always. Then we attempt to insert the new final character to the active point (the root), and find that it is there already. So we end the current step without inserting anything and update the active point to (root,'a',1).

In step 8,, #=8, we append b, and as seen before, this only means we update the active point to (root,'b',2) and increment remainder without doing anything else, because b is already present. However, we notice (in O(1) time) that the active point is now at the end of an edge. We reflect this by re-setting it to (node1,'\0x',0). Here, I use node1 to refer to the internal node the ab edge ends at."

So i'm seeing some contradiction here. Are you setting the current edge always to be whatever edge starts with whatever letter you are considering inserting? Or are you considering only an edge which contains the same substring as the suffix you are considering inserting??

deleted by Bill the Lizard May 18 '12 at 10:50

answered May 18 '12 at 8:00
Alex Ch
11 1

@jogojapan won't see this; you've written it as an "answer" to my question -- you need to add a comment to his answer so that he's notified. – Nathan Ridley May 18 '12 at 9:53

About the implementation and tests at gist.github.com/2373868.

The code works not properly on string 'dedododeeodo$', because in your output suffix tree we cannot find suffix 'do$'. The reason is that the linked node for active node node #3 is not null but node #1 (check the output for iteration 12).

converted to a comment by George Stocker ♦ Aug 13 '12 at 21:12

edited May 16 '12 at 8:31

answered May 15 '12 at 11:54
maruan
65 1 1 7

"We set the active point to (root,'a',1). That means the active point is now somewhere in the middle of outgoing edge of the root node that starts with a, specifically, after position 1 on that edge. "

Then

"We proceed to step 7 by setting #=7, which automatically appends the next character, a, to all leaf edges, as always. Then we attempt to insert the new final character to the active point (the root), and find that it is there already. So we end the current step without inserting anything and update the active point to (root,'a',1).

In step 8,, #=8, we append b, and as seen before, this only means we update the active point to (root,'b',2) and increment remainder without doing anything else, because b is already present. However, we notice (in O(1) time) that the active point is now at the end of an edge. We reflect this by re-setting it to (node1,'\0x',0). Here, I use node1 to refer to the internal node the ab edge ends at."

So i'm seeing some contradiction here. Are you setting the current edge always to be whatever edge starts with whatever letter you are considering inserting? Earlier, the active edge was kept at 'a' when we needed to insert 'ab'. In step 8, the current edge is changed to 'b' instead of keeping it at 'a' when you need to insert 'ab'? But at the same time the 'b' edge is only of length 1 so pointing to position 2 after b doesn't make sense. I'm sure myself and other readers are missing something.

deleted by Bill the Lizard May 18 '12 at 10:50

answered May 18 '12 at 8:13
Alex Ch
11 1

---

how did remainder is 4 at step #10 ?

deleted by ThiefMaster ♦ Aug 14 '12 at 10:28

edited Aug 15 '12 at 1:37

answered Aug 14 '12 at 7:43
bicepjai
452 2 5 20

If you are looking for a variant of suffix tree i.e ternary search tree here is a humble implementation

https://github.com/varunpant/TernaryTree/tree/master/TernaryTree

deleted by owner Sep 30 '13 at 8:45

answered Jun 18 '13 at 20:52

varun
2,164 15 23

An implementation in java can be found at
https://gist.github.com/3355993

deleted by Andrew Barber Aug 25 '13 at 1:55

answered Aug 15 '12 at 4:48

bicepjai
452 2 5 20
